

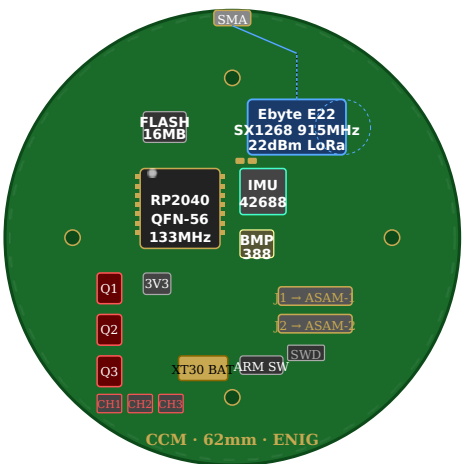
XRIM-117 WYVERN 70mm PTD

PDR-002 Rev A

PCB Layout Diagrams — CCM · ASAM-1 · ASAM-2 | Skylight Industries LLC | 62mm Circular, 2-Layer FR4, ENIG

CCM — Central Command Module

RP2040 · ICM-42688-P · BMP388 · Ebyte E22-900M22S · 3 × Pyro MOSFET · 1S 850mAh LiPo · 62mm circular



| Ref    | Component         | Package      | Function                        |
|--------|-------------------|--------------|---------------------------------|
| U1     | RP2040            | QFN-56       | Primary MCU, 133MHz dual M0+    |
| U2     | W25Q128JVS1Q      | SOIC-8       | 16MB QSPI flash — telemetry log |
| U3     | ICM-42688-P       | LGA-14       | Primary IMU, SPI 32kHz ODR      |
| U4     | BMP388            | LGA-8        | Primary barometer, I2C 200Hz    |
| U5     | Ebyte E22-900M22S | SMD module   | LoRa 915MHz, SX1268, 22dBm      |
| U6     | TLV62569          | SOT-23-5     | 3.3V 600mA buck for logic       |
| Q1-Q3  | IRFZ44N ×3        | TO-220/D2PAK | Pyro MOSFET channels            |
| J1     | JST-GH 8-pin      | Through-hole | ASAM-1 ribbon connector         |
| J2     | JST-GH 8-pin      | Through-hole | ASAM-2 ribbon connector         |
| J3     | SMA edge-mount    | PCB mount    | 915MHz antenna pigtail          |
| J4     | XT30-M            | Through-hole | 1S 850mAh LiPo input            |
| J5     | 2-pin screw term. | Through-hole | Physical arm switch             |
| J6-J8  | 2-pin screw ×3    | Through-hole | Pyro e-match outputs CH1-3      |
| SW1    | TC2030            | SMD 6-pin    | SWD debug + UART boot           |
| LED1-3 | Bicolor LED ×3    | 0805         | Pyro continuity (CH1-3)         |

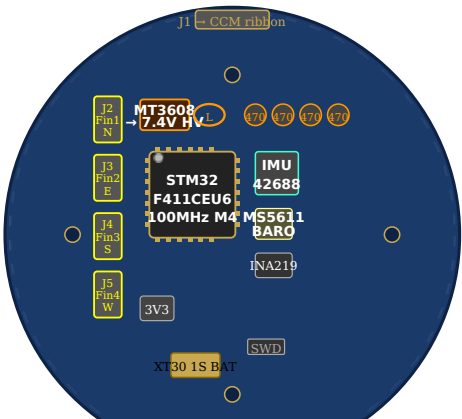
Passives: 100nF+10µF decoupling on each IC power rail. Antenna keepout: 10mm no-copper around SMA. 100Ω series on crystal lines.

SMA connector sits at 12 o'clock board edge. LoRa RF trace from E22 to SMA must be 50Ω controlled impedance (~0.9mm trace width on 1.6mm FR4). Keep 10mm copper-free keepout around antenna pad. Route IMU SPI traces <20mm with matched lengths.

LoRa / RF Pyro MOSFET MCU / logic IC Connector / power Sensor

ASAM-1 — Mid Ring Controller

STM32F411 · ICM-42688-P · MS5611 · 4 × Servo Outputs (Ring 2 mid fins) · MT3608 7.4V boost · 1S 1000mAh LiPo · 62mm circular



| Ref   | Component          | Package      | Function                            |
|-------|--------------------|--------------|-------------------------------------|
| U1    | STM32F411CEU6      | UFQFPN-48    | MCU, Cortex-M4 100MHz FPU           |
| U2    | ICM-42688-P        | LGA-14       | Redundant IMU, SPI 32kHz            |
| U3    | MS5611             | SMD-8        | Redundant barometer, SPI 150Hz      |
| U4    | INA219             | SOT-23-8     | Servo current monitor (7.4V rail)   |
| U5    | TLV62569           | SOT-23-5     | 3.3V 600mA logic buck               |
| U6    | MT3608             | SOT-23-6     | Boost: 1S LiPo → 7.4V HV servo rail |
| L1    | 4.7µH inductor     | 0805         | MT3608 boost inductor               |
| J1    | JST-GH 8-pin       | Through-hole | CCM inter-board ribbon (top edge)   |
| J2-J5 | JR/Futaba 3-pin ×4 | Through-hole | Servo outputs — Ring 2 fins N/E/S/W |

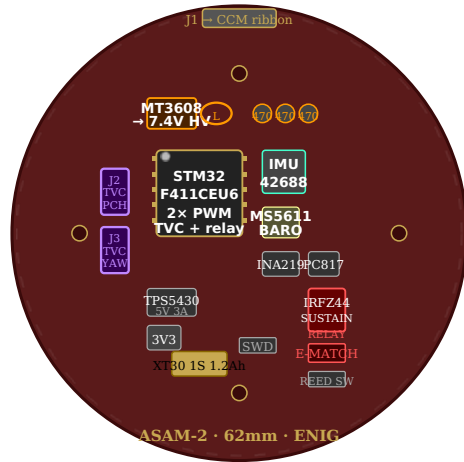
| Ref    | Component             | Package      | Function                            |
|--------|-----------------------|--------------|-------------------------------------|
| J6     | XT30-M                | Through-hole | 1S 1000mAh LiPo input               |
| C1-C4  | 470µF electrolytic ×4 | Radial 6.3mm | Bulk capacitance on 7.4V servo rail |
| C5-C20 | 100nF ceramic ×16     | 0402         | Decoupling on all IC power pins     |
| R1-R4  | 10kΩ ×4               | 0402         | I2C pull-ups SDA/SCL                |
| SW1    | TC2030                | SMD 6-pin    | SWD programming + UART bootloader   |

PWM: TIM1\_CH1-CH4 on PA8-PA11. Servo connector order clockwise from North: J2/J3/J4/J5. 7.4V HV rail separate from 3.3V logic.

MT3608 boost converter requires careful layout: short traces from inductor to IC, large pour on output side, output caps close to servo connector ground pins. 7.4V servo rail is isolated from 3.3V logic rail — ground planes share only at one star point near the input connector.

## ASAM-2 — TVC + Sustainer Controller

STM32F411 · ICM-42688-P · MS5611 · 2× TVC Servo Outputs (jetavane pitch/yaw) · Sustainer Relay · MT3608 7.4V · 1S 1200mAh LiPo · 62mm circular



| Ref   | Component             | Package      | Function                                    |
|-------|-----------------------|--------------|---------------------------------------------|
| U1    | STM32F411CEU6         | UFQFPN-48    | MCU, Cortex-M4 100MHz, 2× PWM TIM outputs   |
| U2    | ICM-42688-P           | LGA-14       | Redundant IMU, SPI 32kHz                    |
| U3    | MS5611                | SMD-8        | Redundant barometer, SPI 150Hz              |
| U4    | INA219                | SOT-23-8     | Servo current monitor (7.4V rail)           |
| U5    | TLV62569              | SOT-23-5     | 3.3V 600mA logic buck                       |
| U6    | MT3608                | SOT-23-6     | Boost: 1S LiPo → 7.4V HV servo rail         |
| U7    | TPS5430               | TO-263-7     | 5V 3A buck (extra MCU headroom)             |
| U8    | PC817 optocoupler     | DIP-4        | Sustainer ignition relay isolation          |
| Q1    | IRFZ44N               | TO-220/D2PAK | Sustainer e-match MOSFET relay              |
| J1    | JST-GH 8-pin          | Through-hole | CCM inter-board ribbon (top edge)           |
| J2-J3 | JR/Futaba 3-pin ×2    | Through-hole | TVC Pitch servo (J2) and TVC Yaw servo (J3) |
| J4    | XT30-M                | Through-hole | 1S 1200mAh LiPo input                       |
| J5    | 2-pin screw term.     | Through-hole | Sustainer e-match output                    |
| J6    | 2-pin header          | Through-hole | Booster eject reed switch                   |
| C1-C3 | 470µF electrolytic ×3 | Radial 8mm   | Bulk cap on 7.4V rail (2 TVC servos)        |
| L1    | 4.7µH inductor        | 0805         | MT3608 boost inductor                       |
| SW1   | TC2030                | SMD 6-pin    | SWD debug + UART boot                       |

TVC servos on TIM2\_CH1 (PA0) and TIM2\_CH2 (PA1). Aft fins are passive (no PWM channels required). Sustainer relay fires on UART command from CCM decoded by STM32 → PC817 → IRFZ44N → e-match.

With passive aft fins, ASAM-2's servo load drops dramatically — only the 2 TVC servos remain on the 7.4V rail. Peak simultaneous stall is  $2 \times 1.5A = 3A$  instantaneous. Three 470µF bulk capacitors are more than adequate to prevent voltage sag during TVC corrections. Place capacitors within 5mm of TVC connector ground pins. IRFZ44N gate must still be driven via PC817 optocoupler for galvanic isolation from CCM logic ground.

■ Sustainer relay ■ TVC servo outputs ■ Boost converter ■ Power connector

XRIM-117 WYVERN PDR-002 Rev A — PCB Layout Reference | Skylight Industries LLC | CONFIDENTIAL | All boards: 62mm circular, 2-layer FR4 1.6mm, ENIG, JLCPCB fabrication